1.

How many outputs are on a BCD decoder?

A. 4 B. 16

C. 8 D. 10

Answer: Option D

2.

In a Gray code, each number is 3 greater than the binary representation of that number.

A. True B. False

Answer: Option B

3.

Use the weighting factors to convert the following BCD numbers to binary.
0101 0011 0010 0110 1000

A. 01010011 001001101000

B. 11010100 100001100000

C. 110101 100001100

D. 101011 001100001

Answer: Option C

4.

Which digital system translates coded characters into a more useful form?

A. encoder B. display

C. counter D. decoder

Answer: Option D

5.

From the following list of input conditions, determine the state of the five output leads on a 74148 octal-to-binary encoder.

$I_0 = 1$	$I_3 = 1$	$I_6 = 1$
$I_1 = 1$	$I_4 = 0$	$I_7 = 1$
$I_2 = 1$	$I_5 = 1$	$EI = 0$

A. GS = L, A_0 = L, A_1 = L, A_2 = H, EO = H

B. GS = L, A_0 = H, A_1 = L, A_2 = L, EO = H

C. GS = L, A_0 = L, A_1 = H, A_2 = L, EO = H

D. GS = L, A_0 = H, A_1 = H, A_2 = L, EO = H

Answer: Option D

6.

What is the function of an enable input on a multiplexer chip?

A. to apply V_{cc}

B. to connect ground

C. to active the entire chip

D. to active one half of the chip

Answer: Option C

7.

The expansion inputs to a comparator are used for expansion to a(n):

A. 4-bit system

B. 8-bit system

C. BCD system

D. counter system

Answer: Option B

8.
What do the mathematical symbols $A < b$ and $A > B$ mean?

A. $A < B$ means A is greater than B. $A > B$ means A is less than B.

B. $A > B$ means A is less than B. $A < B$ means A is greater than B.

C. $A < B$ means A is less than B. $A > B$ means A is greater than B.

Answer: Option C

9.
A basic multiplexer principle can be demonstrated through the use of a:

A. single-pole relay

B. DPDT switch

C. rotary switch

D. linear stepper

Answer: Option C

10.
How many inputs will a decimal-to-BCD encoder have?

A. 4 B. 8

C. 10 D. 16

Answer: Option C

11.
A principle regarding most IC decoders is that when the correct input is present, the related output will switch:

A. active-HIGH

B. to a high impedance

C. to an open

D. active-LOW

Answer: Option D

12. What control signals may be necessary to operate a 1-line-to-16 line decoder?

A. flasher circuit control signal

B. a LOW on all gate enable inputs

C. input from a hexadecimal counter

D. a HIGH on all gate enable circuits

Answer: Option B

13. One multiplexer can take the place of:

A. several SSI logic gates

B. combinational logic circuits

C. several Ex-NOR gates

D. several SSI logic gates or combinational logic circuits

Answer: Option D

14. How many exclusive-NOR gates would be required for an 8-bit comparator circuit?

A. 4 B. 6

C. 8 D. 10

Answer: Option C

15. How many inputs are required for a 1-of-10 BCD decoder?

A. 4 B. 8

C. 10 D. 1

Answer: Option A

16. A BCD decoder will have how many rows in its truth table?

A. 10 B. 9

C. 8 D. 3

Answer: Option A

17. How many possible outputs would a decoder have with a 6-bit binary input?

A. 16 B. 32

C. 64 D. 128

Answer: Option C

18. Most demultiplexers facilitate which type of conversion?

A. decimal-to-hexadecimal

B. single input, multiple outputs

C. ac to dc

D. odd parity to even parity

Answer: Option B

19. The inputs/outputs of an analog multiplexer/demultiplexer are:

A. bidirectional

B. unidirectional

C. even parity

D. binary-coded decimal

Answer: Option A

20. Why can a CMOS IC be used as both a multiplexer and a demultiplexer?

A. It cannot be used as both.

B. CMOS uses bidirectional switches.

Answer: Option B

21. One application of a digital multiplexer is to facilitate:

A. data generation

B. serial-to-parallel conversion

C. parity checking

D. data selector

Answer: Option D

22. The primary use for Gray code is:

A. coded representation of a shaft's mechanical position

B. turning on/off software switches

C. to represent the correct ASCII code to indicate the angular position of a shaft on rotating machinery

D. to convert the angular position of a shaft on rotating machinery into hexadecimal code

Answer: Option A

23.
Why is a demultiplexer called a data distributor?

A. The input will be distributed to one of the outputs.

B. One of the inputs will be selected for the output.

C. The output will be distributed to one of the inputs.

Answer: Option A

24.
What is the status of the inputs S_0, S_1, and S_2 of the 74151 eight-line multiplexer in order for the output Y to be a copy of input I_5?

A. $S_0 = 0, S_1 = 1, S_2 = 0$

B. $S_0 = 0, S_1 = 0, S_2 = 1$

C. $S_0 = 1, S_1 = 1, S_2 = 0$

D. $S_0 = 1, S_1 = 0, S_2 = 1$

Answer: Option D

25.
One way to convert BCD to binary using the hardware approach is:

A. with MSI IC circuits

B. with a keyboard encoder

C. with an ALU

D. UART

Answer: Option A

26.
A microcontroller differs from a microprocessor in that it has several _____ ports and _____

built into its architecture, making it better suited for _____ applications.

A. communication, PROMs, control

B. parallel, logic gates, processing

C. input/output, memory, control

D. data, memory, decoding

Answer: Option C

27.

How is an encoder different from a decoder?

A. The output of an encoder is a binary code for 1-of-*N* input.

B. The output of a decoder is a binary code for 1-of-*N* input.

Answer: Option A

28.

Why is the Gray code more practical to use when coding the position of a rotating shaft?

A. All digits change between counts.

B. Two digits change between counts.

C. Only one digit changes between counts.

Answer: Option C

29.

For the following conditions on a 7485 magnitude comparator, what will be the state of each of the three outputs?

$A_0 = 0$	$B_0 = 1$	$I_A < B = 0$
$A_1 = 1$	$B_1 = 0$	$I_A = B = 1$
$A_2 = 1$	$B_2 = 0$	$I_A > B = 0$
$A_3 = 0$	$B_3 = 0$	

A. $A = B = 0, A < B = 0, A > B = 1$

B. $A = B = 0, A < B = 1, A > B = 0$

C. $A = B = 1, A < B = 0, A > B = 0$

D. $A = B = 0, A < B = 0, A > B = 0$

Answer: Option A

30. When two or more inputs are active simultaneously, the process is called:

A. first-in, first-out processing

B. priority encoding

C. ripple blanking

D. first-in, first-out processing or priority encoding

Answer: Option B

31. A binary code that progresses such that only one bit changes between two successive codes is:

A. nine's-complement code

B. 8421 code

C. excess-3 code

D. Gray code

Answer: Option D

32. Which of the following is not a weighted value positional numbering system:

A. hexadecimal

B. binary-coded decimal

C. binary

D. octal

Answer: Option B

33.
How many inputs are required for a 1-of-16 decoder?

A. 2 B. 4

C. 8 D. 16

Answer: Option B

34.
A truth table with output columns numbered 0–15 may be for which type of decoder IC?

A. hexadecimal 1-of-16

B. dual octal outputs

C. binary-to-hexadecimal

D. hexadecimal-to-binary

Answer: Option A

35.
In a BCD-to-seven-segment converter, why must a code converter be utilized?

A. to convert the 4-bit BCD into 7-bit code

B. to convert the 4-bit BCD into 10-bit code

C. to convert the 4-bit BCD into Gray code

D. No conversion is necessary.

Answer: Option A

36.
How can the active condition (HIGH or LOW) or the decoder output be determined from the logic symbol?

A. A bubble indicates active-HIGH.

B. A bubble indicates active-LOW.

C. A square indicates active-HIGH.

D. A square indicates active-LOW.

Answer: Option B

37. If two inputs are active on a priority encoder, which will be coded on the output?

A. the higher value

B. the lower value

C. neither of the inputs

D. both of the inputs

Answer: Option A

38. A circuit that responds to a specific set of signals to produce a related digital signal output is called a(n):

A. BCD matrix

B. display driver

C. encoder

D. decoder

Answer: Option C

39. How many 74184 BCD-to-binary converters would be required to convert two complete BCD digits to a binary number?

A. 8 B. 4

C. 2 D. 1

Answer: Option C

40. How many select lines would be required for an 8-line-to-1-line multiplexer?

A. 2

B. 3

C. 4

D. 8

Answer: Option B